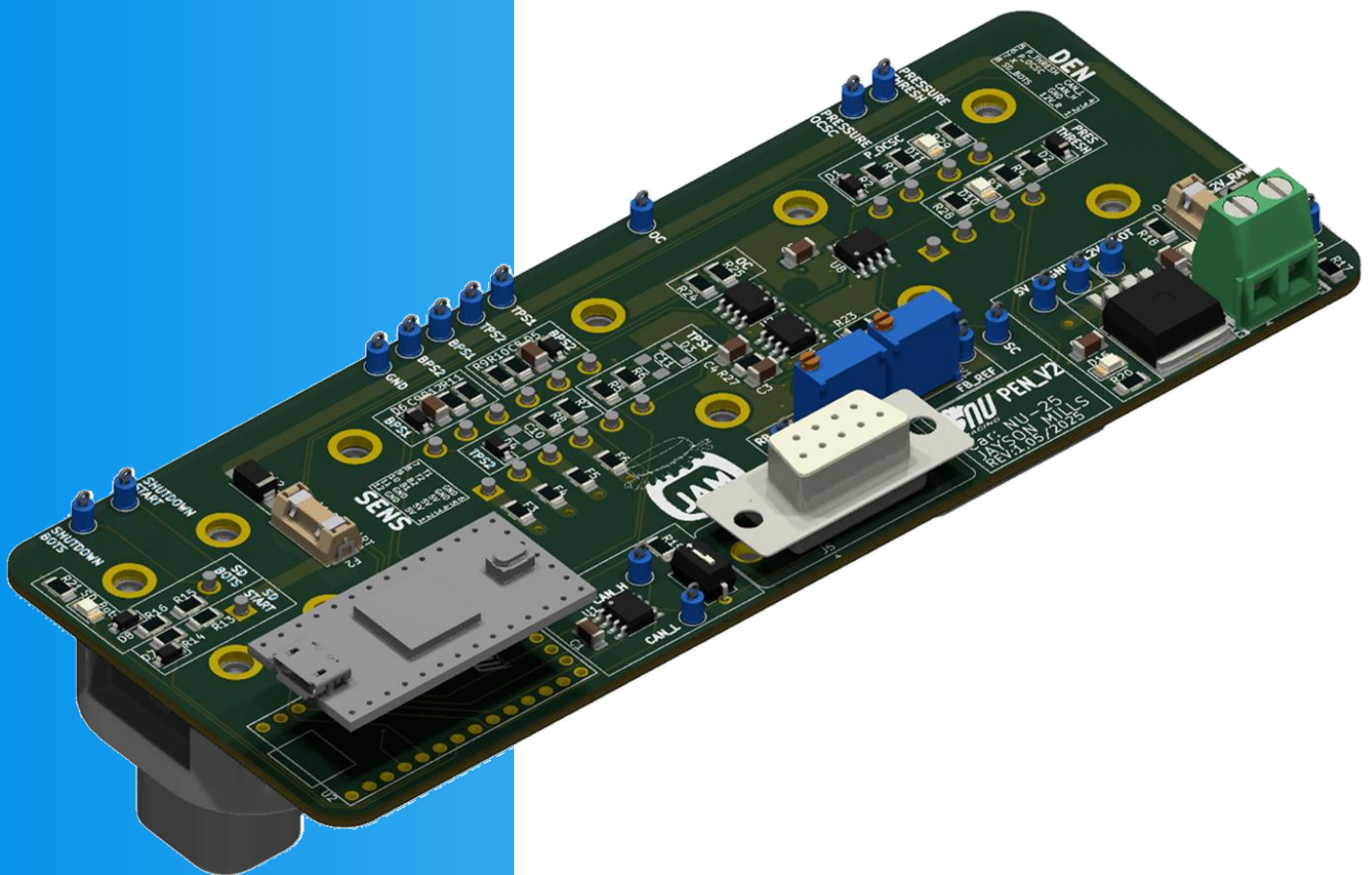




Pedal Electrical Node

DESIGN REPORT



Jayson Mills
2025



Contents

List of Figures	ii
List of Tables	iii
Introduction	1
Node Implementation.....	1
Scope of PEN_V2.....	1
System Design	3
Topology / Schematic.....	3
Standard Circuits.....	3
Shut Down.....	3
Sensors	4
Throttle Linearity	5
Brake Input Filtering.....	6
Final Signal Circuits	7
BSPD	9
Component Selection.....	10
Layout.....	11
Manufacturing and Commissioning.....	13
Manufacturing Lessons	13
Tuning	13
Integration	13
Adaptation to NU24.....	13
Shutdown Circuit Issues.....	14
PEN Integration on Car.....	14
Unit Test Checklist/ Commissioning Procedure.....	15
Bill of Materials	16
Conclusions and Recommendations	17
References	18
Appendix	19
PEN Sensor testing.....	19
PEN PCB and Node Related Rules [6].....	27
Schematic.....	30
Trace Layout.....	31
3D Renders.....	32

List of Figures

Figure 1: PEN Implementation in the LV Topology	1
Figure 2: Flowchart of Signal Processing Through PEN_V2	3
Figure 3: a, SHUTDOWN_START Fusing and Protection. b, SHUTDOWN_BOTS Pinout	4
Figure 4: Shutdown Pulldown For Teensy	4
Figure 5: Sensor Loom Pinout	5
Figure 6: Testing Pedal Sensors (Left), PEN_V1Sensor Input Dynamics (Right)	5
Figure 7: [3]Typical Zener Breakdown Characteristics for BZX84C_V_	6
Figure 8: Voltage Divider + RC Filter Hand Calculations	7
Figure 9: Voltage Divider + RC Filter Simulation	7
Figure 10: a, TPS Pulldown and Clamping. b, BPS Pulldown and Filtering New to PEN_V2	8
Figure 11: OCSC Schematic	9
Figure 12: Threshold Schematic.....	10
Figure 13: 2024 PEN_V1 Layout.....	11
Figure 14: 2025 PEN_V2 Layout.....	11
Figure 15: 2025 PEN_V2 Final Board View.....	12
Figure 16: Unit Testing of PEN Following the Checklist	13
Figure 17: Unit Test Checklist/ Commissioning Procedure.....	15
<i>Figure 1: PEN V1 PCB with important connections.</i>	<i>19</i>
<i>Figure 2: Raw TPS output test points</i>	<i>20</i>
<i>Figure 3: Teensy pin out for required signals</i>	<i>20</i>
<i>Figure 4: TPS2 (Zener) Expected result. Teensy (purple) max 2.6V.....</i>	<i>21</i>
<i>Figure 5: TPS1 (no diode) expected outcome. Teensy (purple) max 3.3V.....</i>	<i>21</i>
<i>Figure 6: TPS1 pulldown using signal generator. 0.72-4.03 V input with NO zener.</i>	<i>22</i>
<i>Figure 7: TPS1 pulldown using signal generator. 0.72-4.03 V input WITH zener.</i>	<i>22</i>
<i>Figure 8: [3]Typical zener breakdown characteristics for BZX84_V_.....</i>	<i>22</i>
<i>Figure 9: BPS Test points (see figure 1 for layout)</i>	<i>23</i>
<i>Figure 10: BPS no capacitor.</i>	<i>23</i>
<i>Figure 11: BPS with capacitor.</i>	<i>24</i>
<i>Figure 12: Oscilloscope measuring BPS1 whilst PEN is in the car with a driver.</i>	<i>25</i>
<i>Figure 13: Max brake pressure recorded during competition brake test</i>	<i>25</i>
<i>Figure 14: BPSX with 0.1uF capacitor simulation vs bench test (no latency)</i>	<i>25</i>



List of Tables

Table 1: Sensor Output Ranges..... 5

Table 2: BOM 16

Table 1: TPS Range Results..... 21

Table 2: BPS Results..... 24

Introduction

This report provides a detailed account of the design, manufacture and commissioning of the Pedal Electrical Node (PEN) Printed Circuit Board (PCB) version 2 (PEN_V2). During semester 1 of 2025, the task to develop a new version of the PEN_V1, created by Eliza O'Donnell was scoped to adapt the design for NU Racing's NU25 FSAE competition platform.

The creation of the PCB benefited greatly from the support of Jacob Lukes, Malcolm Sidney, and Alec Chapman.

Node Implementation

The PEN is a constituent of the NU25 nodal LV (Low Voltage) topology. The node's PCB facilitates the reading of the brake pressure sensors (BPS1 & BPS2) and throttle position sensors (TPS1 & TPS2). The shutdown circuit originates from the node with SHUTDOWN_START, and it accommodates the Brake Over Travel Switch (SHUTDOWN_BOTS). The PEN also implements the brake pressure half of the BSPD (Brake System Plausibility Device) circuit. The FSAE rules relating to the PEN system are available in the appendix under "PEN PCB and Node Related Rules".

The PEN connects to the Dash Electrical Node (DEN) which provides power, CANBUS communication lines, and relays important PEN signals to the rest of the car, as demonstrated in Figure 1. For more information pertaining to the LV topology (of NU24), see Alec Chapman's report "ENGG4200 Mechatronic Lead 2024" [1] (available at: NU Teams Discord -> NU racing -> fyp-docs -> Alec Chapman ENGG4200 Mechatronic Lead 2024).

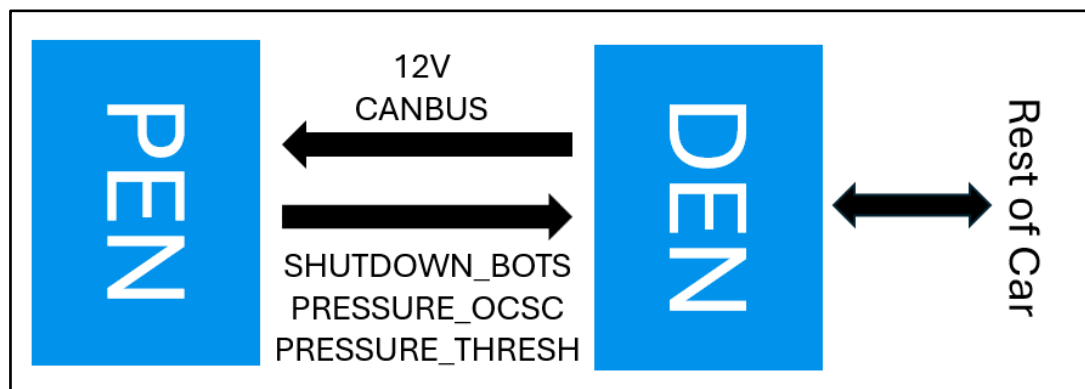


Figure 1: PEN Implementation in the LV Topology

Scope of PEN_V2

The PEN_V2 iteration retained the pre-existing circuit schematic with improvements to the analogue signal treatment, while overhauling the physical layout to accommodate the new pedal box design and housing. The new design required the PCB to be vertical behind the pedals, with connectors running perpendicular to the car's direction of travel. This was a change from the previous pedal box layout where the PCB lay horizontal underneath the pedals. The PCB revision provided an opportunity to enhance multiple aspects of the design:

- The non-linearity of TPS1 due to the Zener diode was investigated and fully understood, with the PEN_V1 temporary solution formally validated through testing.
- A filter was added to the brake sensor inputs to reduce the noise on the signal. The filter design was researched and developed to satisfy the Teensy input protection requirements while adding no voltage drop to the brake signals. This avoids changing OCSC (Open Circuit, Short Circuit) and pressure threshold voltage requirements.



- Serviceability was improved by grouping all 'to car' connectors on the back side of the board, with all other circuitries being accessible on the outward facing side. Test points were populated with accessibility in mind, and effective use of the silkscreen improved commissioning and testing.
- Trace routing was improved with more deliberate placement of functional circuit groups which eliminated vias and reduced the length of analogue traces. This reduced noise and error on critical signals.

System Design

Topology / Schematic

As the most significant change to the schematic was the filtering on the BPS signals, the treatment of signals through the PCB was unchanged from PEN_V1. Figure 2 demonstrates the flow and processing of signals across the PCB.

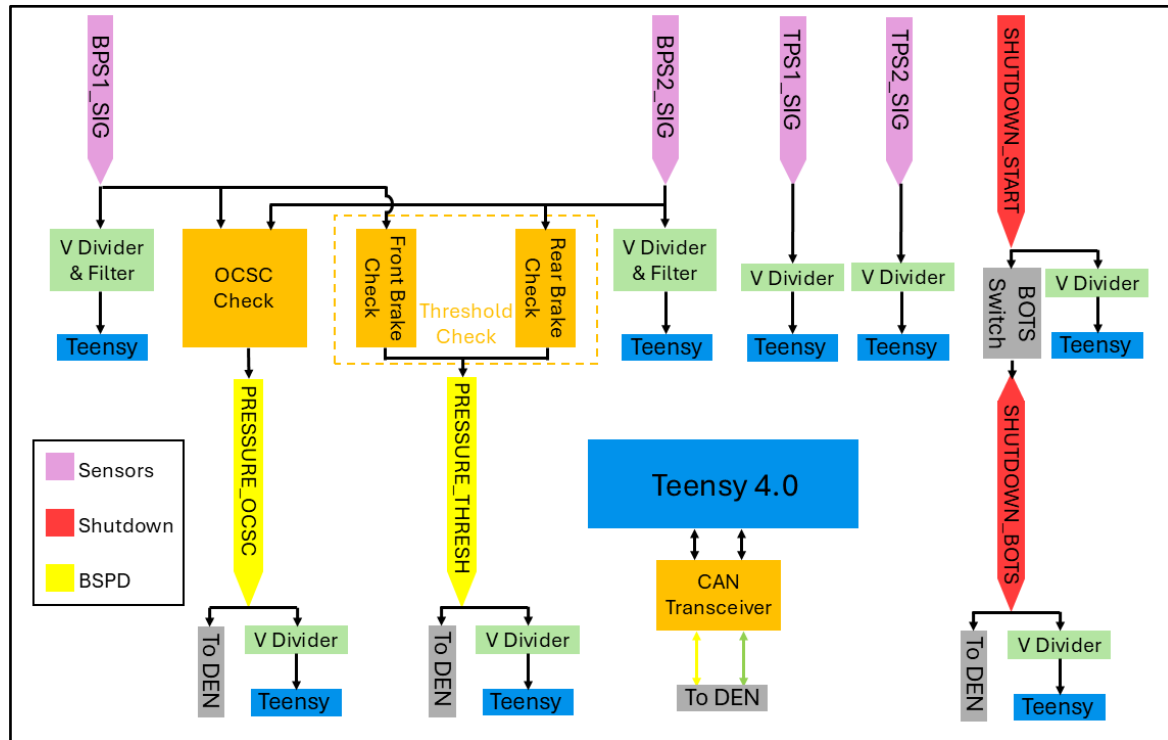


Figure 2: Flowchart of Signal Processing Through PEN_V2

Standard Circuits

The voltage regulation and CAN transceiver circuits are homologated with the NU Teams Standard Circuits KiCad Library and therefore will not be explained. For more detail on the standard circuits and components, refer to "Daniel_Iveson_FYP_Report" [2], section 5.2 (NU Teams Discord -> NU Racing -> fyp-docs).

Shut Down

The 12 V shutdown circuit originates from the PEN. SHUTDOWN_START is fused and polarity protected from 12_RAW as per any other power source and for rules compliance. This circuit was inherited from PEN_V1, shown in Figure 3a. Shutdown BOTS is a normally closed toggle switch activated by the over extension of the brake pedal. When BOTS becomes an open circuit, the shutdown circuit is broken, stopping the car. BOTS is connected using a 2-pin DT connector with the pinout demonstrated by Figure 3b.

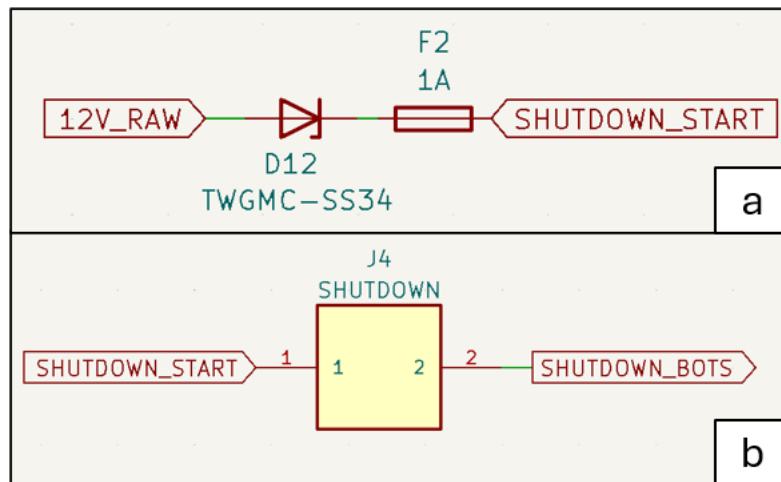


Figure 3: a, SHUTDOWN_START Fusing and Protection. b, SHUTDOWN_BOTS Pinout

The state of the shutdown circuit is logged to the Teensy, and then transmitted over CAN. The shutdown voltage is stepped down using a divider to bring 12 V to below 3.3 V to be safe for Teensy measurement. A 3.3 V Zener diode clamps the voltage to ensure the Teensy is safe in the event of a greater than 12 V input voltage to the divider. The input protection circuit shown in Figure 4 was adapted from PEN_V1, with the addition of a status LED for BOTS for serviceability and design continuity with other PCBs shutdown indicators. The logging of the state on the microcontroller is only for monitoring as the shutdown circuit must be an unbroken analogue loop.

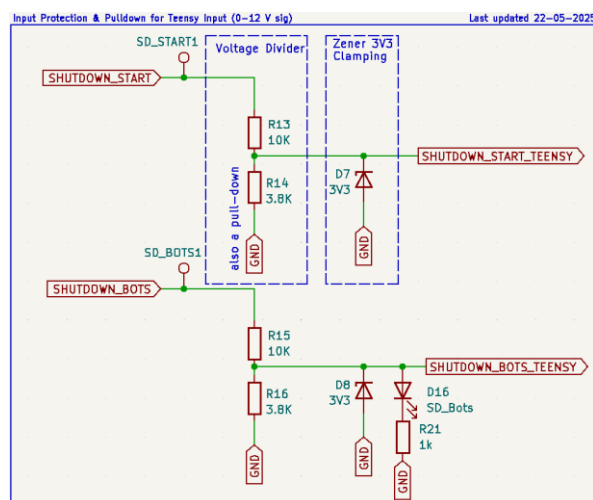


Figure 4: Shutdown Pulldown For Teensy

Sensors

The throttle pedal is an OEM part from a 2013 VW Golf MK6 (part ID: 1k2 721 503 S BA), and the brake pressure sensors are Honeywell Heavy Duty Pressure Transducers (part ID: MLH02KPSB06A). A loom that adapts the different sensor connections to the PCB DT connector has the pinout displayed in Figure 5. The sensors require 5V and are individually fused to prevent hardware faults from impacting the PCB (the fuses were a PEN_V2 addition). To specify a fuse size, the sensors were individually powered with a power supply and used while monitoring the current draw. A fuse size of 0.1 A was selected. This could be improved by using measurement equipment suited for the application or contacting Volkswagen for a sensor datasheet.

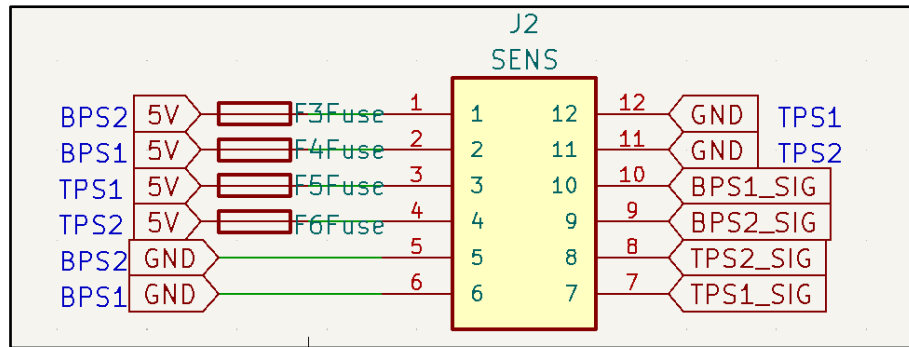


Figure 5: Sensor Loom Pinout

PEN_V1 had noise and linearity issues with the Teensy pulldown and protection circuits for the sensors. The temporary fix for the old version was to not populate the TPS1 clamping diode and to add capacitors to the brake inputs without a permanent footprint. These changes were anecdotal and not validated through calculations or testing so an experiment on PEN_V1 was conducted to gain insight for the design of PEN_V2. For the full experiment details, see “PEN Sensor Testing” within the appendix.

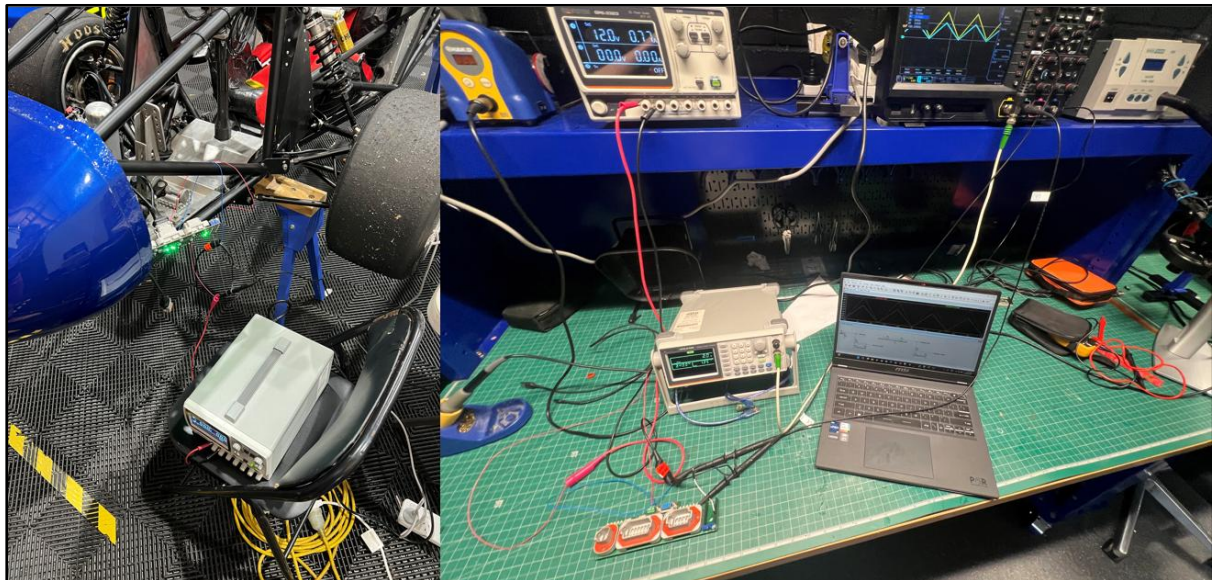


Figure 6: Testing Pedal Sensors (Left), PEN_V1 Sensor Input Dynamics (Right)

Throttle Linearity

In completing this experiment, the operational voltage output ranges of the sensors were found to be not as expected. TPS1 and TPS2 did not operate with the same ranges demonstrated by Table 1. Further investigation revealed this discrepancy was accounted for in the PEN Teensy code with an offset and scaling, however this was undocumented knowledge.

Table 1: Sensor Output Ranges

		TPS1	TPS2	BPS1	BPS2
Sensor	Output	0.7-4.0	0.5-2.2	0.7- ~ 3.1	0.7- ~3.3
	Voltage Range (V)				

With these ranges, a function generator was utilised to replicate transient use of the pedals during bench testing PEN_V1. This validated the LTspice models developed and revealed the cause of TPS1 being the only signal losing linearity at high throttle application. The low current application of the clamping 3.3 V Zener diode results in the zener breakdown voltage remaining in the ‘soft knee’ area of the BZX84C3V3 diode. This only impacts TPS1 due to the higher voltage post voltage divider being above ~2.2 V. This is where the soft knee characteristics begin, as displayed in Figure 7. For further analysis, see “PEN Sensor Testing” within the appendix.

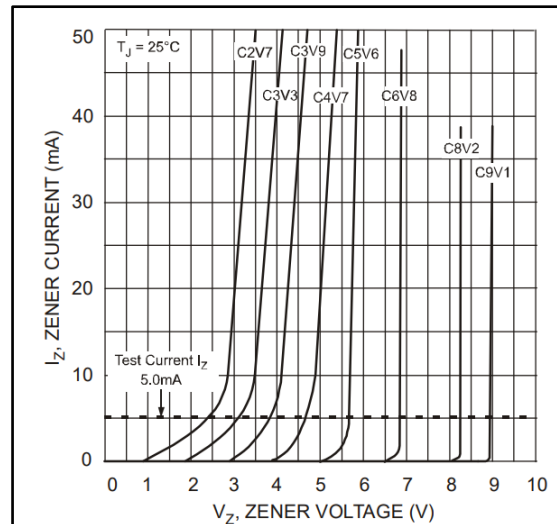


Figure 7: [3]Typical Zener Breakdown Characteristics for BZX84C_V_

Brake Input Filtering

In 2024, it was identified using the MoTeC data that the brake pressure signals contained noise. The interim fix was to solder 0.1 μF capacitors between the raw inputs and GND underneath the DT connectors. PEN_V2 improved this by developing a voltage divider and RC low-pass filter input pulldown circuit. During experimentation (see appendix for “PEN Sensor Testing”), the sensor noise was not able to be measured resulting in an unknown noise frequency. To specify a desired cutoff frequency, the pre-existing filter on the BSPD was checked to roughly estimate an applicable frequency. Figure 8 demonstrates the calculation required to design the filter. The design considered the operating window of the Teensy, any voltage drops impacting OCSC and threshold calibrations, and avoiding a large capacitance that would result in input delay. A cut off frequency of 2.4Hz was reached using standard component values. This design was then validated using a simulation as shown in Figure 9.

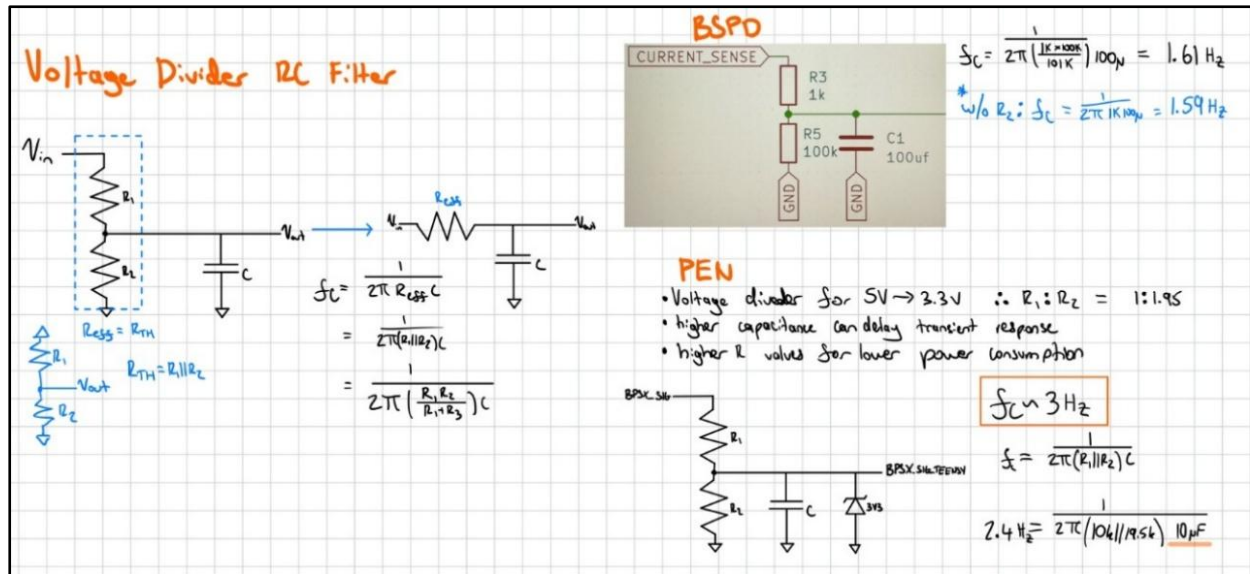


Figure 8: Voltage Divider + RC Filter Hand Calculations

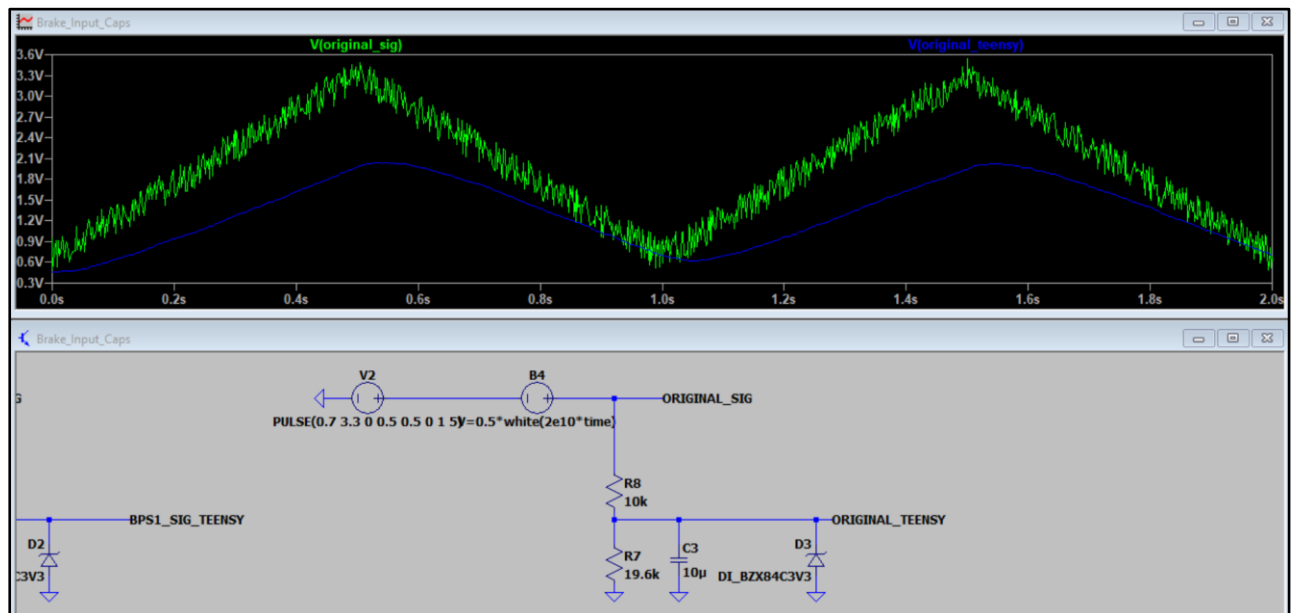


Figure 9: Voltage Divider + RC Filter Simulation

Final Signal Circuits

The recommendations of the research detailed the implementation of the TPS configurations displayed in Figure 10a. Empty capacitor footprints were also added for parity with the brake filters as a redundancy in the event it is found necessary. Figure 10b demonstrates the use of the low-pass filter using:

$$f_c = \frac{1}{2\pi \left(\frac{R_1 R_2}{R_1 + R_2} \right) C}$$

where $f_c = 2.4 \text{ Hz}$.

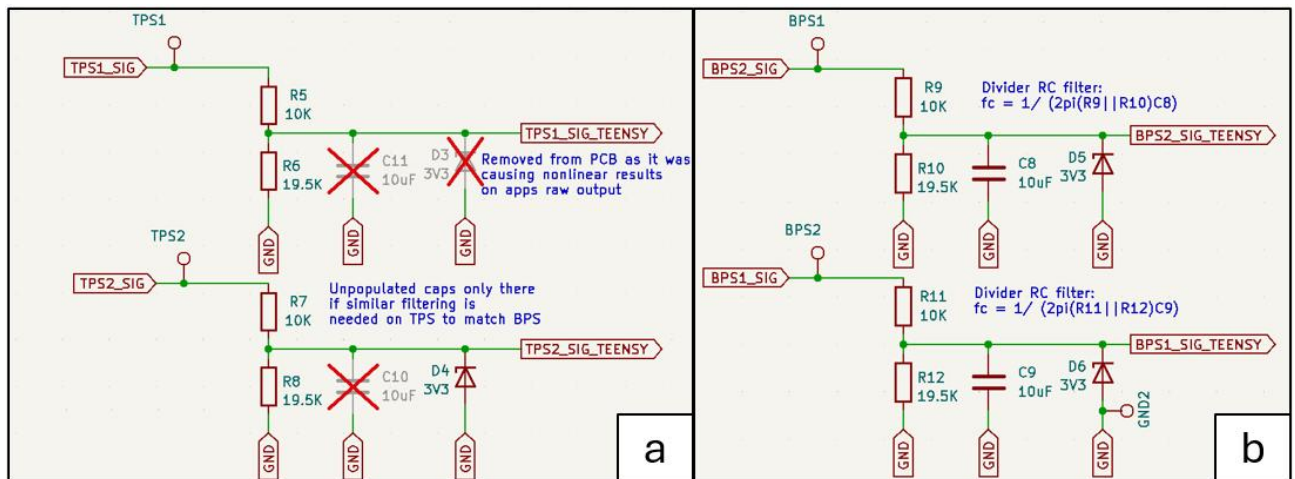


Figure 10: a, TPS Pulldown and Clamping. b, BPS Pulldown and Filtering New to PEN_V2

BSPD

The PEN implements the brake pressure half of the BSPD circuit. The main logic circuit located on the BSPD breakout board required two binary signals from the PEN, `PRESSURE_OCSC` and `PRESSURE_THRESHOLD`. The entire BSPD circuit was inherited from `PEN_V1`.

OCSC checks the sensor outputs for hardware faults and outputs an OK High Signal (OKHS) with high logic indicating a healthy. It does this by generating two reference voltages SC and OC using a divider to be representative of voltages outside the working range of the sensor. These voltages are 4.54 V and 0.45 V respectively as shown in Figure 11. Both BPS signals are checked against these reference voltages using LM393 comparators which set the output low when range is exceeded. The OKHS logic is set using a pull-up resistor and a buffer op-amp isolates the `PRESSURE_OCSC` output and indicator light from the circuit.

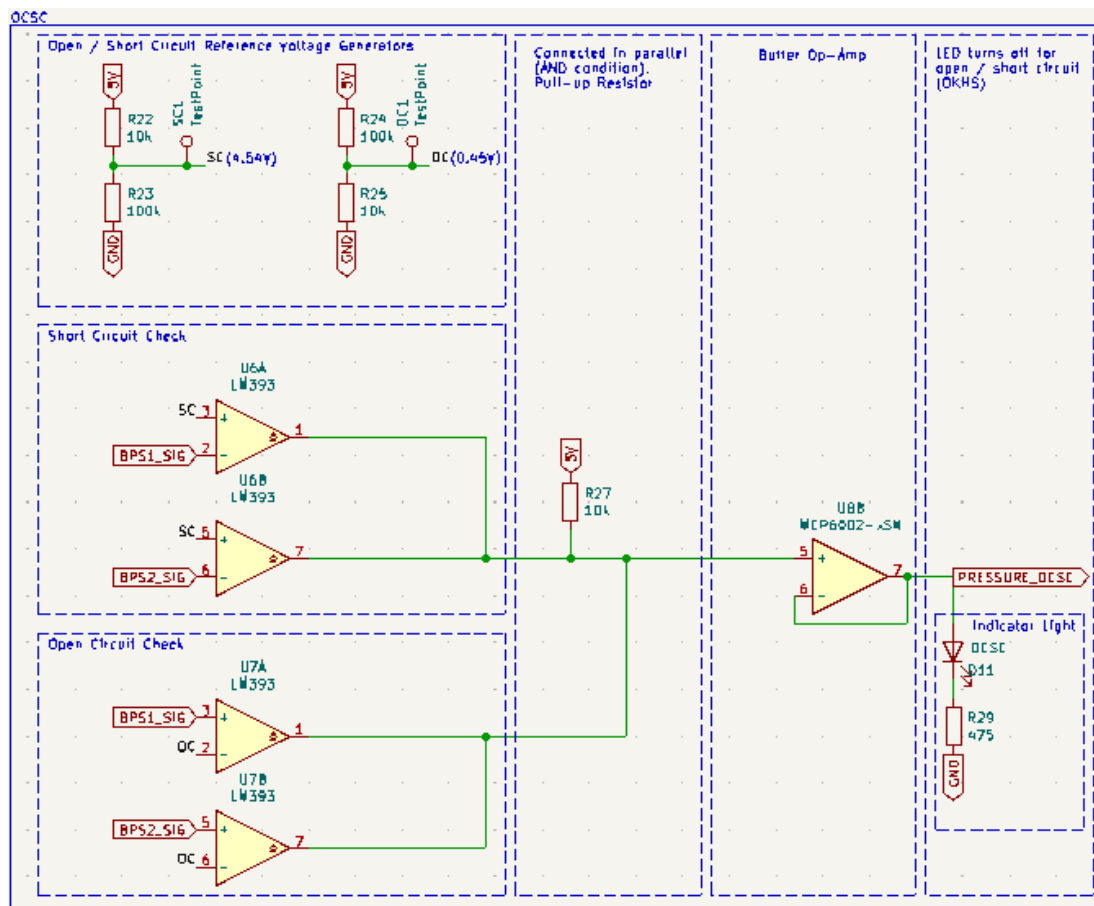


Figure 11: OCSC Schematic

The threshold section of the BSPD circuit checks both front and rear brake sensors for a voltage reflective of a demand for hard braking, demonstrated by Figure 12. The reference voltage can be set for each sensor individually using a 10 kΩ potentiometer which are then fed into LM393 comparators. Similar to OCSC, the output is pulled up to 5 V because the signal is grounded when pressure exceed the threshold.

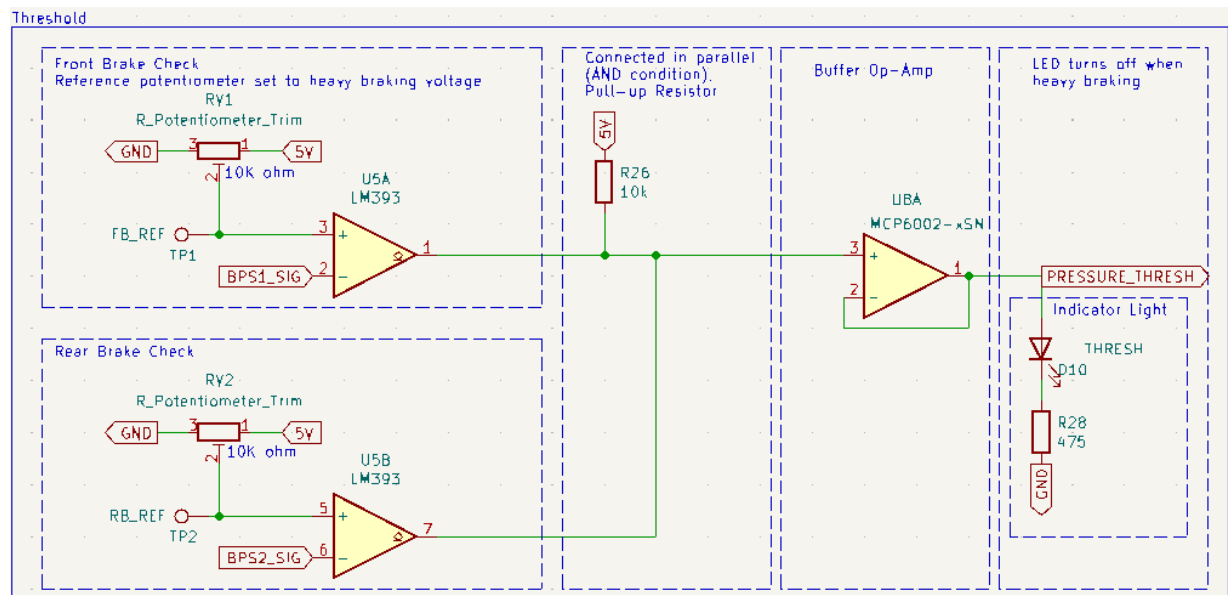


Figure 12: Threshold Schematic

Component Selection

Standard components stocked in the NU Teams MCHA Lab (Mechatronics Lab) were used where possible.

PTC fuses – Instead of using the nano fuse holders found on the voltage regulation and shutdown start circuits, the sensors are fused using surface mount PTC self-resetting fuses. This reduced the footprint size to 1206 and to improve serviceability.

Trimmer Potentiometer - To tune the front and rear brake reference voltages, a 25 turn, top adjustment 10 kΩ potentiometer is used. The high turn count allows for precise adjustment of the resistance and reference voltage.

Layout

With the overhaul of the mechanical pedal box design for NU25, the PEN layout was completely redesigned. The intent of the change was to fit the design into the new space requirements but also to address weaknesses in the PEN_V1 design. V1 optimised for close component grouping and centralisation of the test points as depicted in Figure 13. This compromised trace routing by increasing via usage on analogue lines and reduced the ability to keep sensor traces at similar lengths and to differentially route can bus lines. Since V1 was housed beneath the pedal box, serviceability was not prioritised as the board is inaccessible.

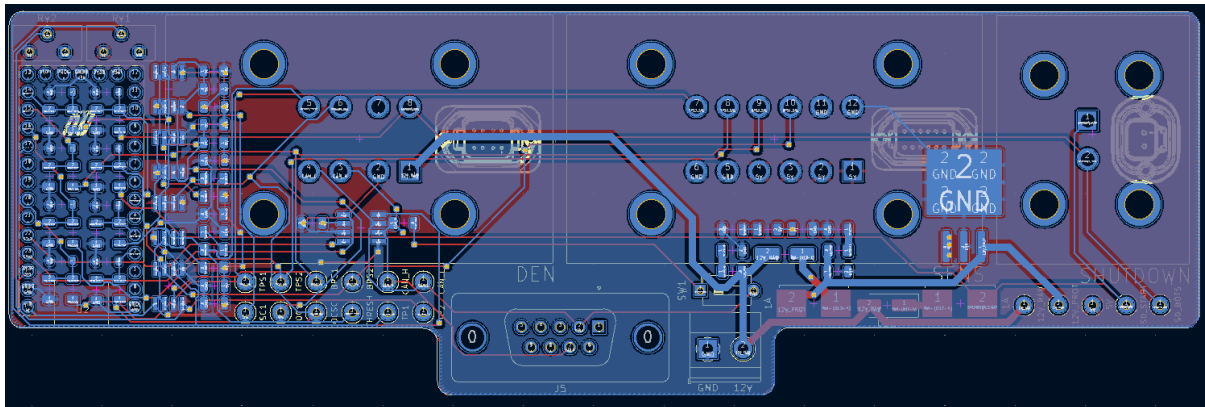


Figure 13: 2024 PEN_V1 Layout

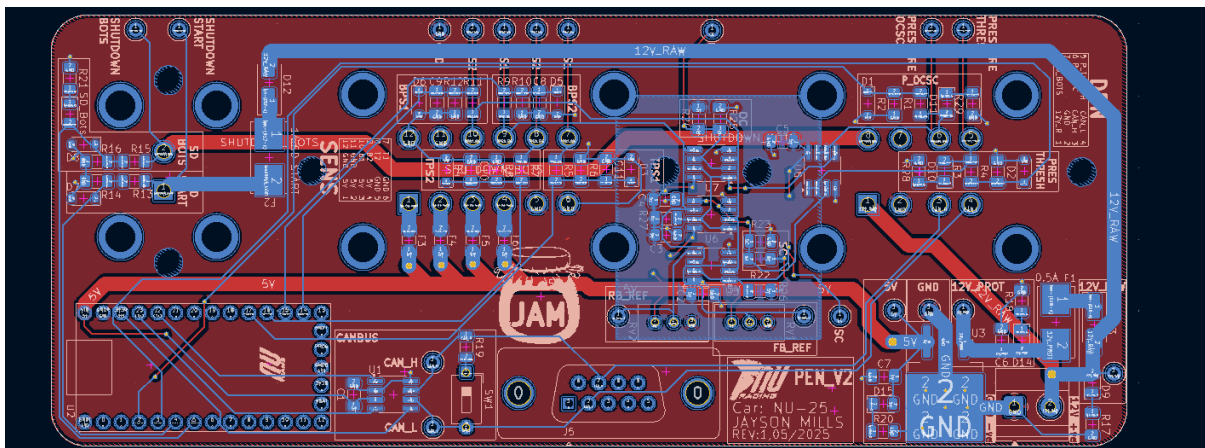


Figure 14: 2025 PEN_V2 Layout

PEN_V2 improved the design by considering the following:

- One sided PCB – With the DT connectors placed first and reflecting the same configuration as PEN_V1, all other components were placed on the opposite side of the board. This better utilised the space behind the DTs and allows the board to be mounted to car on the DT side while still being exposed for testing and servicing with the removal of the housing lid.
- Functional group placement – Illustrated in Figure 14, each functional block was placed next to the corresponding connectors and DT pinouts. Test points were also populated adjacent to the origin of the corresponding signals reducing trace complexity and ease of distinction between neighbouring points.
- Analogue Traces – No vias were used on analogue signals and TPS and BPS lines were kept at similar lengths. This reduces noise and disturbance in the Teensy readings.

- Shutdown – Shutdown trace width was increased to 2mm as per NU Racing requirements. Reducing resistance and voltage drops along the shutdown circuit is a priority.
- CAN – CAN lines were kept at an identical length with no use of vias.
- Serviceability - Serviceability was greatly improved with the clear use of silkscreen blocking and labelling to separate between distinct circuits which can be observed in Figure 15.

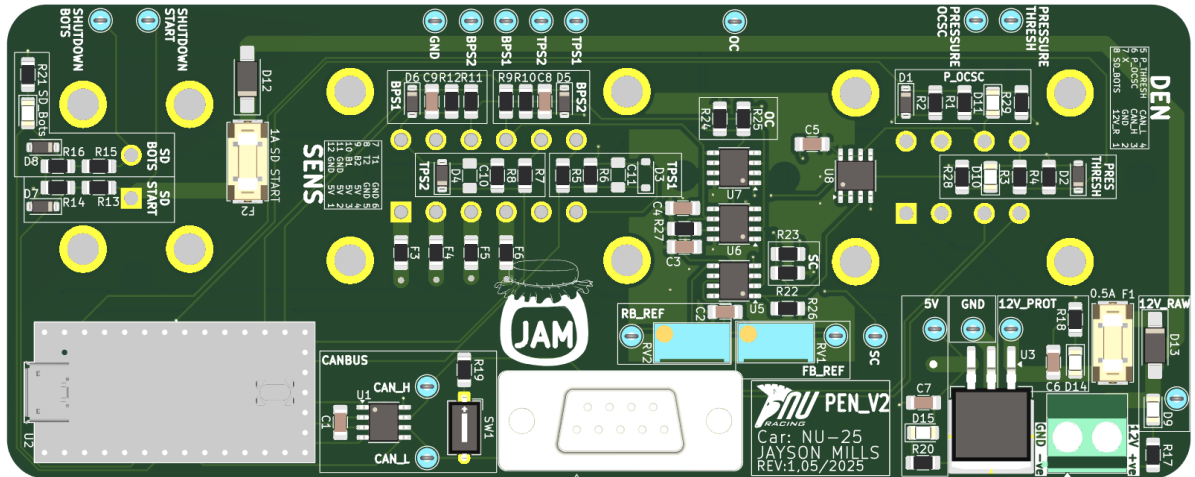


Figure 15: 2025 PEN_V2 Final Board View

Manufacturing and Commissioning

With the design finalised and submitted for manufacture with JLCPCB, a unit test and commissioning procedure was planned, provided in Figure 17. Once the blank PCB was delivered, the procedure was executed. Minimal manufacturing issues occurred, and all unit tests demonstrated expected outcomes, the process is illustrated in Figure 16.

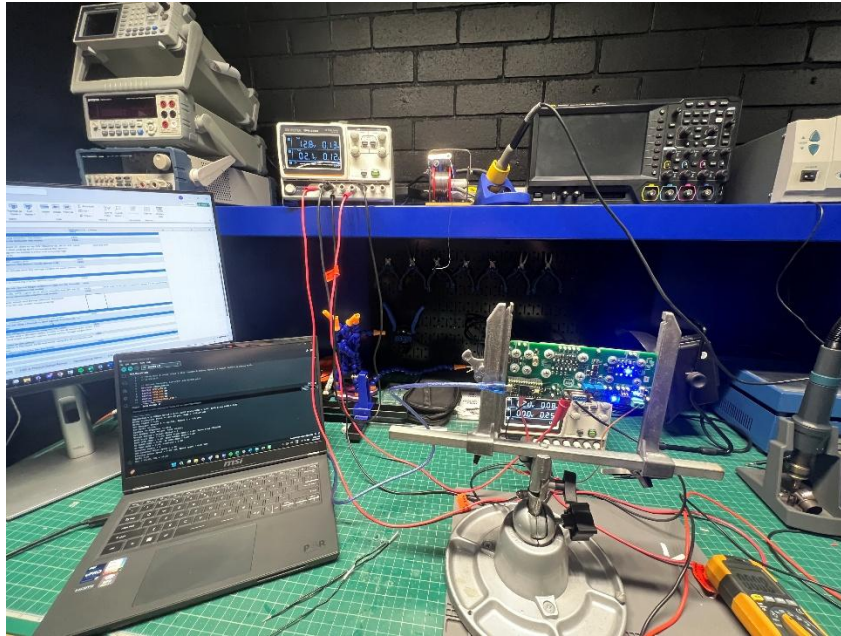


Figure 16: Unit Testing of PEN Following the Checklist

Manufacturing Lessons

During manufacturing, it was found that the 5 V regulator was positioned too close to the commissioning screw terminal for adequate soldering. The polymer screw terminal body was at risk of melting when applying heat to the GND pad of the regulator. Post manufacture, the population of certain beaded test point loops was reconsidered. Any test point that did not require the use of an oscilloscope hook probe had the test loop unpopulated to ease the use of multimeter probing. As a result, only the analogue sensor test points were populated whereas any VDC or logic points remained as an exposed copper pad.

Tuning

To save time tuning on the car, which is critical during LV integration timelines, the front and rear brake pressure reference voltages were tuned using bench testing. To accomplish this, MoTeC data from the previous PEN was reviewed to determine at what brake pressures `PRESSURE_THRESH` was being tripped. MoTeC logs the pressure in bar to the nearest whole number, which limited the precision of the exercise. Once converted to kPa, this was then converted to a voltage using the PEN code's analogue read calculation as a guideline. 10 bar threshold pressure equated to approximately 0.8 V at the sensors. The reference voltages were preliminarily tuned to 0.8 V.

Integration

Adaptation to NU24

NU Racing's 2025 electronics were tested using the 2024 competition platform prior to its decommissioning. To accommodate PEN_V2 on NU24, Joshua Garbutt created a temporary 3D printed

housing that mounted the PCB to the chassis members beside the pedal box. Oliver Dowzer created a new BOTS DT loom to account for the extra distance between the switch and the PCB.

Shutdown Circuit Issues

There was a delay between bench testing the PEN and validating it on NU24 due to various issues along the shutdown circuit. These issues were troubleshooted by Jacob Lukes, the author, Alec Chapman, and the entire mechatronics team.

Interpose Relay:

Situated on the CEN (Central Electrical Node), an interpose relay boosts the shutdown signal to 12 V before it enters the accumulator. This is required as the shutdown circuit runs the length of the car roughly twice and through most PCBs which results in large voltage drops along the line. With the addition of the new CEN, the shutdown voltage dropped below the minimum switching voltage of the relay. As a result, the accumulator was not receiving the required healthy ~12 V high shutdown signal and could not enter HV (High Voltage) state. The initial fix for this issue was to use a lower voltage rated interpose relay (changing from 12 V to 5 V able to switch 200% of supply voltage).

Discharge Relay:

With the interpose relay fix the car was then able to switch to HV. However, the accumulator would fail to pre-charge in a sufficient amount of time resulting in the accumulator shortly leaving HV. It was found that the normally closed discharge relay on the TSAL_Discharge board located on the HIP PCB (Human Interface Panel) failed to open leaving the 5 k Ω discharge resistor connected in series with the tractive system HV. This caused the pre-charge failures.

The discharge relay [4] (positioned prior to the interpose relay) is rated for 12 V switching voltage but was experimentally found to have a minimum of approximately 7.6 V. The new 5 V interpose relay [5] changed the interposed relay coil resistance from 960 Ω to 167 Ω , which allowed for more current draw and higher voltage sag from the non-fixed voltage supply of the shutdown circuit. As a result, the half of the shutdown circuit prior to the boosting of the interpose, sagged below the switching open voltage of the discharge relay.

With this discovery, the interpose was reverted to the original 12 V relay and methods to reduce voltage drop along the shutdown were pursued. Zak Lobko decreased the resistance of the new CENs shutdown traces by adding supporting wires in parallel. It was also discovered that the Brake Over Travel Switch produced an abnormally high voltage sag and was determined to be the main issue.

BOTS:

It was discovered that when creating the new loom, the switch had been connected using an incorrect pinout. The toggle switch is classed as a single pole single through (SPST) yet has 3 spade terminals. The new loom connected to the first and third pins (intended for 12cV and GND) whereas the second pin (the pole) was left disconnected. Intended use case is to connect to the pole for the signal then either 12V or GND to determine pull up or pull-down behaviour. To close the shutdown circuit, SHUTDOWN_START and SHUTDOWN_BOTS must be connected to adjacent terminals on the switch.

PEN Integration on Car

With the cancellation of multiple track days due to mechanical manufacturing timelines and mechatronic issues, as of the publishing date of this report, the PEN_V2 has not been testing during a track day. It has however been successfully tested on the car demonstrating throttle and brake application as well as BSPD functionality.

Unit Test Checklist/ Commissioning Procedure

The procedure in Figure 17 is located in the NU Teams GitHub repository at Racing-NU25->PEN->PCB->PEN V2->PEN V2_DDR-> Unit Test Checklist.

Stage	Function	Description	Verification Method	Steps	Result	Notes
Before Assembly	QUALITY CONTROL	PCB Quality Control	Inspection	Check the PCB is has been manufactured to our expected quality	PASS	
POWER						
Voltage Regulation and Protection	POWER DISTRIBUTION	Power and Ground are isolated	Test	Use a multimeter to read that the resistance between power and ground is an open loop	PASS	
Before Assembly	ASSEMBLY	Solder commissioning input and 12V_RAW status LED and test point	Inspection	Visually inspect solder joints, measure continuity between net nodes then supply 12V at the input and observe 12V_RAW LED on	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder reverse polarity protection, fuse and status LED + TP	Inspection	Visually inspect solder joints, measure continuity between net nodes then supply 12V at the input and observe 12V_PROT LED on	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder 5V regulator circuit and status point	Inspection	Visually inspect solder joints, measure continuity between net nodes then supply 12V at the input and observe 5V LED on	PASS	
During Assembly - Unpowered	POWER DISTRIBUTION	Check all supply voltages	Test	Supply the commissioning port with 12V and measure all of the supply voltages to be true.	PASS	
TEENSY						
During Assembly - Unpowered	ASSEMBLY	Solder teensy pins	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Powered	Teensy power	Check power	Inspection	Measure teensy supply voltage	PASS	
SHUTDOWN						
During Assembly - Unpowered	ASSEMBLY	Solder shutdown fuse and diode	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder Input Protection & Pulldown for Teensy Input	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Powered	SHUTDOWN	Validate SHUTDOWN	Test	Supply the commissioning port with 12V, measure ST_Start to be 12V. Observe SD_BOTS LED on only when a closed circuit is between SD start and SD BOTS (simulating the switch). Measure SD_Bots as 12V and both teensy signals as below 3.3 but still a teensy high (zener diode soft knee may limit this to ~2.2 V)	PASS	
SENSORS						
During Assembly - Powered	ASSEMBLY	Solder PTC fuses	Inspection	Supply 12V, Measure 5V at the 4 TPS and BPS supply pins	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder the 4 input filters checking for unpopulated components.	Inspection	Supply each sensor input with 5v and measure the teensy inputs below 3.3V	PASS	
During Assembly - Powered	SENSORS	Replicate PEN V1 Sensor Testing results with singal generator and oscilloscope	Analysis	Using the report as a guide, replicate the results with the voltage ranges for each sensor	PASS	
PRESSURE						
During Assembly - Unpowered	ASSEMBLY	Solder voltage dividers and input protection for OCSC and Tresh	Test	Supply Pressure OCSC and Thresh with 5v checking 3.3v for teensy inputs	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder OCSC circuit	Inspection	Solder reference voltages, comparitors etc. Do not forget protection caps for lcs. Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Powered	OCSC	Test OCSC fault for each sensor	Test	Observe indicator LED on and PRESSURE_OCSC signal at 5V when healthy and off + 0V for faulted	PASS	
During Assembly - Unpowered	ASSEMBLY	Solder threshold check circuit with protection caps.	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
During Assembly - Powered	Threshold	Set pressure front and back reference voltages with the potentiometers. (PEN V1 values should be checked to avoid unnecessary tuning while on the car)	Inspection	Supply critical brake sensor voltages above and below reference. Measure PRESSURE_THRESH and LED status to be Low under "heavy braking"	PASS	
CAN						
During Assembly - Unpowered	ASSEMBLY	Solder can transeiver circuit	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
Post Assembly	CAN	Read CAN data	Demonstration	Upload PEN code to teensy, use can king + kvaliser to read signals from teensy. i.e. heartbeat, pressures and apps	PASS	
BENCH TESTING						
During Assembly - Unpowered	ASSEMBLY	Solder DT connectors	Inspection	Visually inspect solder joints, measure continuity between net nodes	PASS	
Post Assembly	Code integration	Scaling checks	Demonstration	Using the teensy serial monitor and a variable power supply on the dt sensor pins, check the scaling of the throttle and brake pressure outputs. (check sensor voltage ranges from the testing report)	PASS	
Final Bench Test	Code integration	BOTS	Demonstration	Demonstrate bots being read on the teensy	PASS	
Final Bench Test	Code integration	Thresh	Demonstration	Repeat the PRESSURE_THRESH & PRESSURE_OCSC test while using the DT connectors and reading the teensy values.	PASS	
Final Bench Test	CAN messages	Check outputted values	Demonstration	Repeat the prior 2 tests while reading the can bus traffic	PASS	
INTEGRATION						
Integration Test	POWER DISTRIBUTION	Plug DEN DT into PEN and turn on car LV	Demonstration	Observe the board receive power from the car	PASS	
Integration Test	SD_BOTS	Connect the BOTS loom to PEN	Demonstration	Demonstrate the switch deactivating the LED and monitor the serial channel on the TEENSY or the can read out for BOTS	PASS	
Integration Test	SENSORS	Connect sensor loom	Demonstration	With HV disabled. Test the pedals reading correct values over CAN with correct range	PASS	
Integration Test	BSPD, APPS TRAIL	Test and tune pressure Thresh, observe Pressure OCSC off when sensors not plugged in	Demonstration	Apply pressure to the brake and check threshold. Ensure apps trail is activate before BSPD when on both pedals.	PASS	
Integration Test	APPS GATED	Spin the wheels	Demonstration	Turn on HV, activate RTD, test spinning the wheels and t braking.	PASS	

Figure 17: Unit Test Checklist/ Commissioning Procedure

Bill of Materials

Table 2: BOM

Part	Value	Designator	Footprint	QTY Per PCB	QTY Needed	TA Stock	Ordered	Footprint Fits?	DataSheet
Capacitor	0.1uF	C1, C2, C3, C4, C5, C7	1206	6	30			YES	
Capacitor	10uF	C10, C11, C8, C9	1206	4	20			YES	
Capacitor	0.33uF	C6	1206	1	5			YES	
Screw Terminal		COMMISSIONING_PWR1	TerminalBlock_Phoenix_MKDS-1.5-2_1x02_P5.00mm_Horizontal	1	5			YES	
Zener Diode	3V3	D1, D2, D3, D4, D5, D6, D7, D8	D_SOD-123	8	40			YES	
LED	Blue	D10, D11, D14, D15, D9	1206	5	25			YES	
LED	Red	D16	1206	1	5			YES	
Diode	TWGMCS-SS34	D12, D13	D_SMA_Handsoldering	2	10			YES	
Fuse Holder	(0.5A, 1A)	F1, F2	Fuseholder_Littelfuse_Nano2_154x	2	10			YES	
PTC Fuse	0.1A	F3, F4, F5, F6	1206	4	20	0	25	YES	https://www.belfuse.com/media/datasheets/products/circuit-protection/ds-cp-02cj-series.pdf
Resistor	10K	R1, R11, R13, R15, R22, R25, R26, R27, R3, R5, R7, R9	1206	12	60			YES	
Resistor	19.5K (19.6k)	R10, R12, R2, R4, R6, R8	1206	6	30	0	50	YES	
Resistor	3.8K	R14, R16	1206	2	10			YES	
Resistor	1K	R17, R18, R21	1206	3	15			YES	
Resistor	120	R19	1206	1	5			YES	
Resistor	500 (499)	R20	1206	1	5	0	50	YES	
Resistor	100k	R23, R24	1206	2	10			YES	
Resistor	475	R28, R29	1206	2	10			YES	
Potentiometer	10k	RV1, RV2	Potentiometer_Bourns_3296W_Vertical	2	10	2	8	YES	https://www.bourns.com/docs/Product-Datasheets/p36.pdf
Switch	DSIC01THGET	SW1	SW_DIP_SPSTx01_Slide_6.7x4.1mm_W7.62mm_P2.54mm_LowProfile	1	5			YES	
CAN Transiever	TJA1051T-3	U1	SOIC-8_3.9x4.9mm_P1.27mm	1	5			YES	https://www.nxp.com/docs/en/data-sheet/TJA1051.pdf
Teensy	Teensy4.0	U2	Teensy4.0-NU-Teams	1	1			YES	https://mm.digkey.com/Volume0/opasdata/d220001/medias/docus/329/DEV-15583_Web.pdf
12-5V regulator	LM7805_DPAK	U3	TO-263-3_TabPin2	1	5			YES	https://www.htckorea.co.kr/Datasheet/Voltage%20Regulator/LM78xx.pdf
Dual Comparitor	LM393	U5, U6, U7	SOIC-8_3.9x4.9mm_P1.27mm	3	15			YES	https://www.sikoric.com/Data/sikormicro/upload/file/20250219/LM393.pdf
Dual Op-Amp	MCP6002-xSN	U8	SOIC-8_3.9x4.9mm_P1.27mm	1	5			YES	https://ww1.microchip.com/downloads/en/DeviceDoc/MCP6001-1R-1U-2-4-1-MHz-Low-Power-Op-Amp-DS20001733L.pdf
DT Receptical	SENS	J2	DT15-12PA	1	1			YES	https://www.te.com/commerce/DocumentDelivery/DDEController?Action=srchtrv&DocNm=DT15-12PX&DocType=Customer+Drawing&DocLang=English&PartCntxt=DT15-12PA&DocFormat=pdf
DT Receptical	DEN	J3	DT15-08PA	1	1			YES	https://www.te.com/commerce/DocumentDelivery/DDEController?Action=showdoc&DocId=Customer+Drawing%7FDT15-08PX-XXXX%7FA%7Fpdf%7FEnglish%7FENG_CD_DT15-08PX-XXXX_A.pdf%7FDT15-08PA
DT Receptical	SHUTDOWN	J4	DT15-2P	1	1			YES	https://www.te.com/commerce/DocumentDelivery/DDEController?Action=srchtrv&DocNm=DT15-2P&DocType=Customer+Drawing&DocLang=English&PartCntxt=DT15-2P&DocFormat=pdf
Dsub Socket	DE9_Socket	J5	DSUB-9_Socket_Vertical_P2.77x2.84mm_MountingHoles	1	1			YES	
Throttle Pedal	1k2721 503 S BA	N/A	2013 VW GOLF MK6 Accelerator Pedal	1					
Pressure Sesonrs	MLH02KPSB06A	N/A	Honeywell Heavy Duty Pressure Transducers	2					https://octopart.com/datasheet/mlh02kpsb06a-honeywell-40012085
BOTS Switch	12V Illuminated Toggle RED	N/A	N/A	1					https://www.jaycar.com.au/12v-illuminated-toggle-red/p/ST0580

Conclusions and Recommendations

The new iteration of the PEN was successful in all testing undertaken. However, it is pending the application to the car at the next track day at the time of publication. It demonstrated the ability to supply the car with throttle and brake data, check for brake pedal over travel, and facilitated the pressure half of the BSPD circuit.

Another version of the PEN PCB could utilise low-pass filters on the brake signals prior to the Teensy voltage divider as this would eliminate noise at the OCSC and Threshold circuits too. It must be kept in mind that adding filters before this stage could introduce a voltage drop that would necessitate a recalibration of both circuits reference voltages. Throttle sensor zener diodes could be investigated with possible fixes for the soft knee non-linearity issues being to identify a better clamping method or diode specification that reduces the effect. It may also be possible to utilise zener diodes rated above 3.3 V but get stuck at the low current soft knee characteristics closer to 3.3 V.

Future iterations of the PCB must ensure sufficient communication with the pedal box engineer is maintained. This ensures proper design intent with respect to connector placement, board dimensions, and part placement. It should be noted this becomes difficult if the design cycles do not align.

References

- [1] A. Chapman, "ENGG4200 Mechatronic Lead 2024," NU Racing, Callaghan, 2024.
- [2] D. Iveson, "Daniel Iveson FYP Report," NU Racing, Callaghan, 2024.
- [3] Diodes Incorporated, "Digikey," February 2018. [Online]. Available: <https://www.diodes.com/assets/Datasheets/ds18001.pdf>.
- [4] Sensata Technologies, "Sensata D Series Reed Relay Datasheet," 2024. [Online]. Available: <https://www.cynergy3.com/sites/default/files/sensata-reed-relay-d-series-datasheet.pdf>.
- [5] OMRON Corporation, "Ultra-miniature, Highly Sensitive," [Online]. Available: https://omronfs.omron.com/en_US/ecb/products/pdf/en-g5v_1.pdf.

Appendix

PEN Sensor testing

PEN Sensor Testing

Abstract

This document outlines the testing method and results of measuring the output signals of the throttle position sensors (TPS1 & TPS2) and brake pressure sensors (BPS1 & BPS2) which are located within the pedal box and connected to the Pedal Box Electrical Node (PEN).

Aim

To gain insight into the signal characteristics of the raw analogue inputs to the PEN board which will aid defining the desired input shaping in the new PEN v2 schematic.

Equipment

- NU24 LV systems
- Pedal box & PEN_V1
- Oscilloscope
- Wave form generator
- 12V Power supply (If it is not desirable to utilise the entire LV system the PEN could be powered independently via the commissioning ports).

Procedure

Set up:

- Remove the pedal box from car and the PEN from its enclosure.
- Plug all connectors back into the PEN. The DT12 connector is most relevant as it is associated with all the sensors.
- Bolt the pedal box back into the car to allow for sufficient force to be applied to the pedals.
- Connect the oscilloscope probe GND to the screw terminal GND.

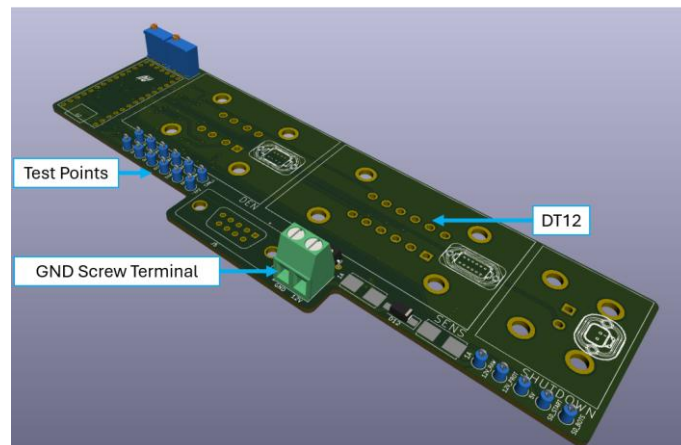


Figure 18: PEN V1 PCB with important connections.

TPS1 & TPS2:

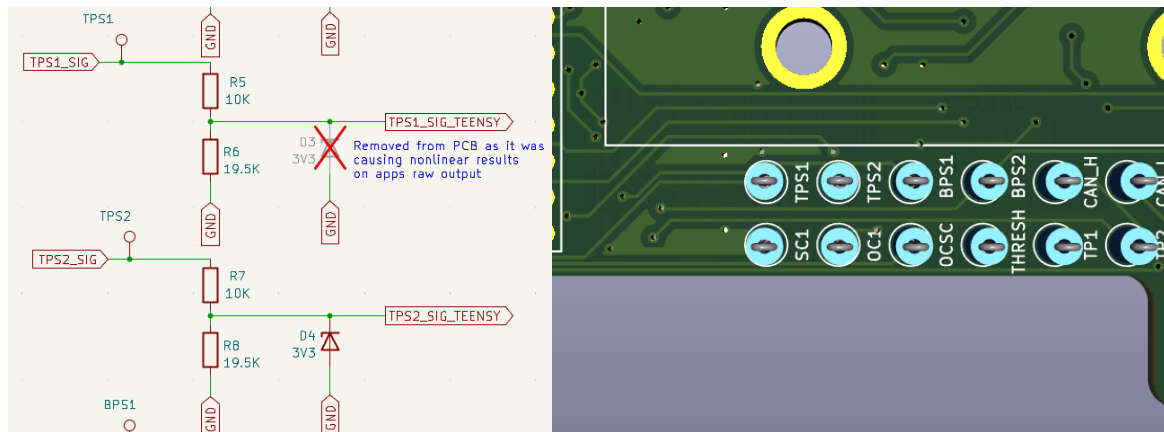


Figure 19: Raw TPS output test points

- Test TPS1 & TPS2 voltage output test (for test points see figure 2) range by checking the zero and full pedal travel positions.
- Using this min and max voltage, set up the PEN with a wave form generator inputting a signal to the TPS test points to represent throttle input. A triangle wave can be used to replicate the driver applying and getting off the throttle. Using two channels on the oscilloscope, compare the generated wave input to the post processed TPSX_SIG_TEENSY (see figure 3 for pinout). The teensy can be removed to access the signals underneath.
- TSP1 Zener diode is unpopulated, observe the linearity of TSP1_SIG_TEENSY at the end of pedal travel (as TPS1_Sig approaches max). Use two probes to observe the input verse output signal.
- TPS2 Zener is populated therefore TSP2_SIG_TEENSY should lose linearity as throttle reaches maximum. Verify this with two probes.

- 14 - BPS1_SIG_TEENSY
- 15 - BPS2_SIG_TEENSY
- 16 - TPS2_SIG_TEENSY
- 17 - TPS1_SIG_TEENSY

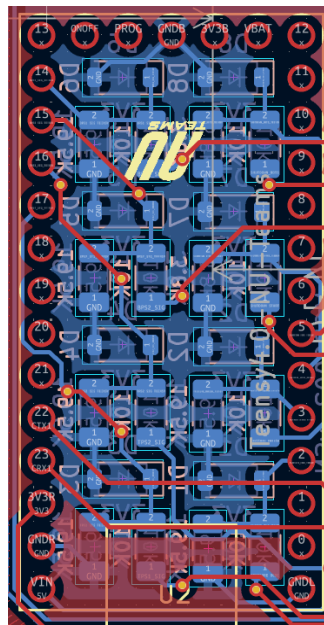


Figure 20: Teensy pin out for required signals

TPS Expected Results (Assuming 0-5V sensor range on LTspice):

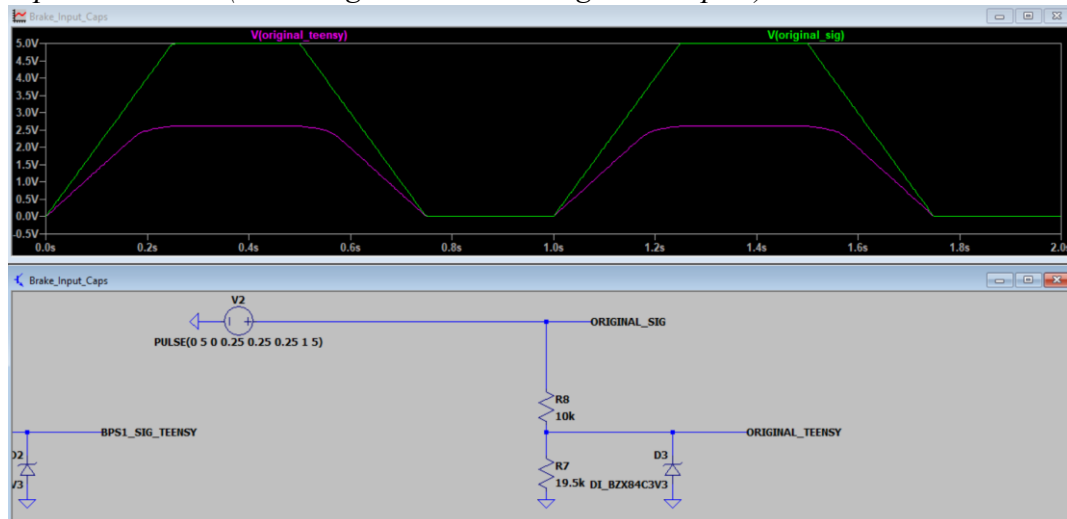


Figure 21: TPS2 (Zener) Expected result. Teensy (purple) max 2.6V

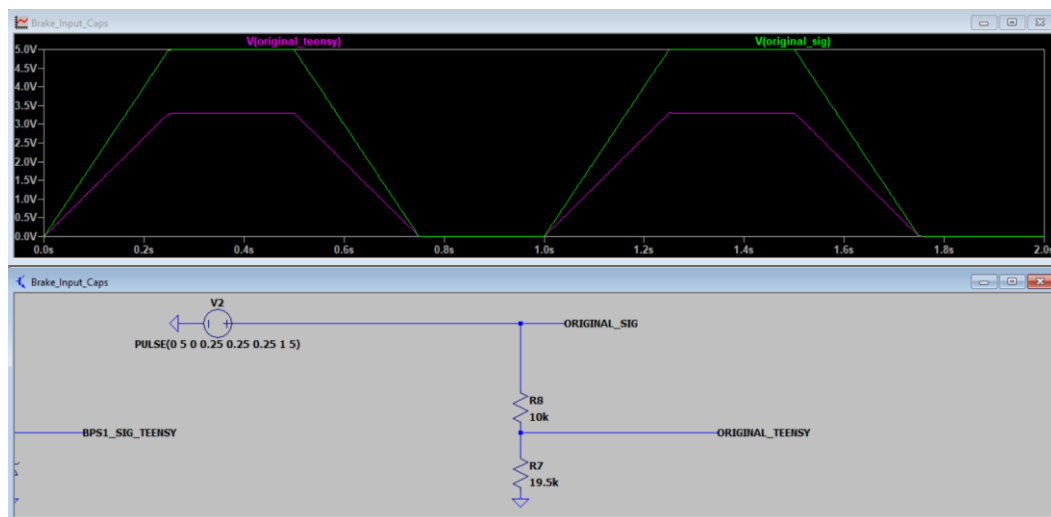


Figure 22: TPS1 (no diode) expected outcome. Teensy (purple) max 3.3V

TPS Results:

Table 3: TPS Range Results

	TPS1	TPS2	APPS out RAW
Sensor Output Voltage Range	0.72-4.03V	0.5-2.2V	0.61-3.115
Post processed Teensy signal range	0.49-2.6 V	0.3-1.45	

To recreate and validate the findings of last years team in repesct to the non-linearity of TPS1 when the clamping zener diode was populated, the TPS1 sensor signal was run through both a non zener (figure 6) and zener (figure 7) configuration.

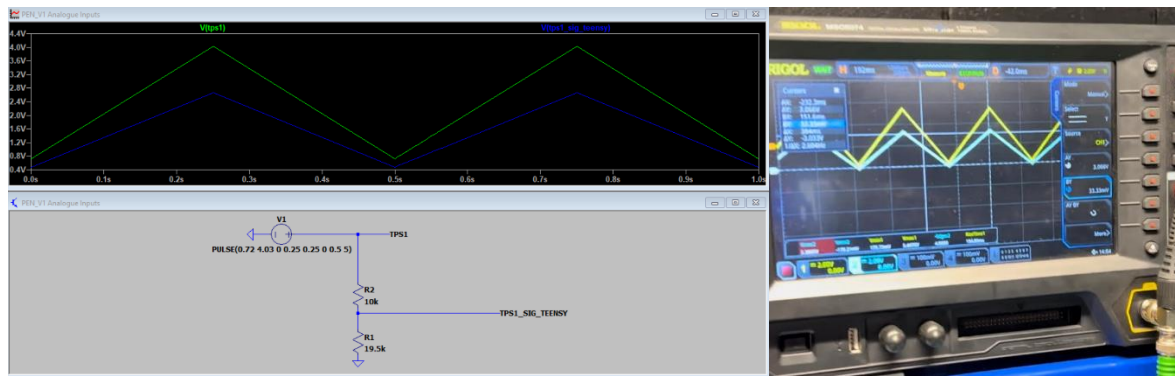


Figure 23: TPS1 pulldown using signal generator. 0.72-4.03 V input with NO zener.

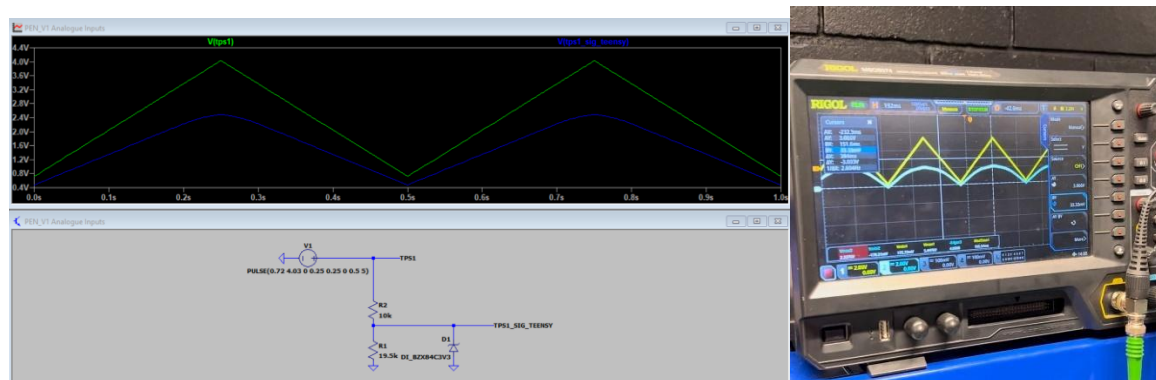


Figure 24: TPS1 pulldown using signal generator. 0.72-4.03 V input WITH zener.

Figure 7 establishes why the TPS1 zener was unpopulated for NU24. The nonlinearity is due to the soft knee characteristics of the 3.3 V zener diode. The low current of the sensor signal means clamping on the voltage begins much earlier than intended resulting in the max voltage getting stuck as the zener begins to let reverse current flow at voltages lower than the desired breakdown (see figure 8 for data sheet zener breakdown).

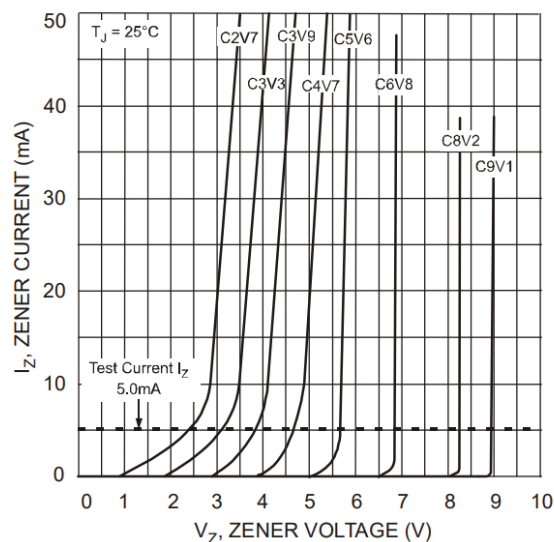


Figure 25: [3]Typical zener breakdown characteristics for BZX84_V_

BPS1 & BPS2

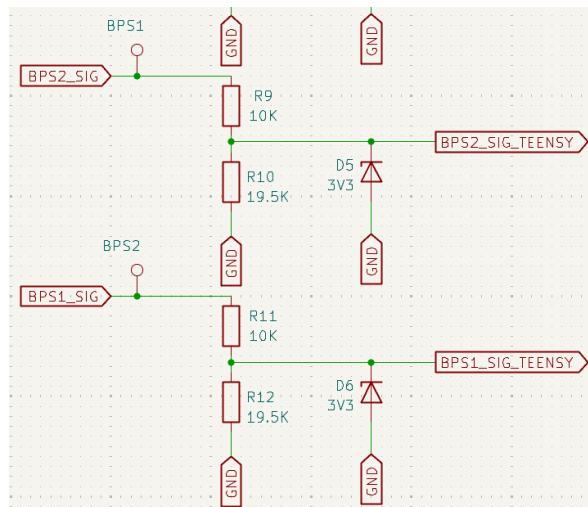


Figure 26: BPS Test points (see figure 1 for layout)

- Similar to TPS1&2, test BPS1 & BPS2 voltage output range.
- Observe transient response and noise level with input level changes.
- Check BPSX_SIG_TEENSY linearity at high pressure.
- Replicate LTspice results using signal generator
- Repeat tests with both versions of the PEN, one with capacitors added on, one without capacitors

BPS Expected Results (LTspice):

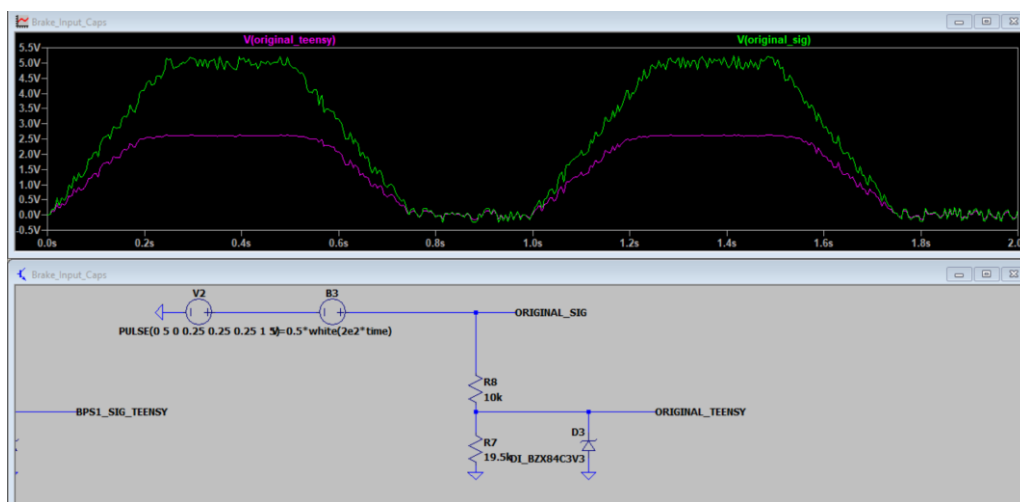


Figure 27: BPS no capacitor.

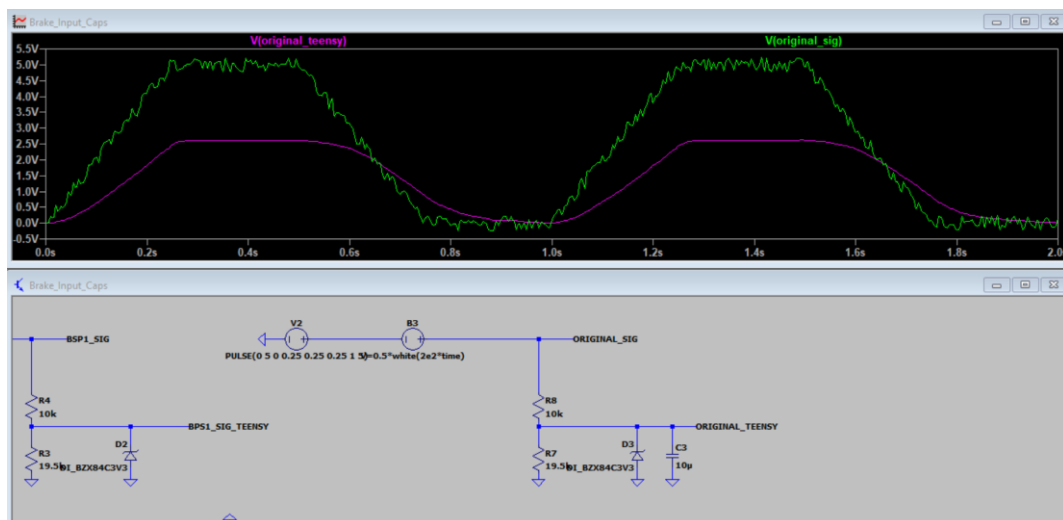


Figure 28: BPS with capacitor.

BPS Results

Table 4: BPS Results

	BPS1	BPS2
Sensor Output Voltage range*	0.7- ~ 3.1V	0.7- ~3.3V
Noise & Transient Response	The noise was not able to be replicated or observed during the test even with high voltage on in the car. However, a different method might need to be developed where the teensy is used to graph the analogue input voltage whilst the PEN is in the car. It was also found that the capacitor was 0.1uF whereas the pre-experiment (figures 10 & 11) simulations used 10uF which resulted in a delayed response but can now be ignored as this is incorrect.	
Zener Linearity & Soft knee	As the post divider voltages are below 2.1V, the signals are unaffected by the zener diode similar to TPS2.	

*max output was dependent on the driver's ability to apply force. 3.1 V was the limit before excessive stress was being applied to the pedal box components causing flexing. Under normal conditions this is an appropriate maximum however it should be noted that higher values can be generated.

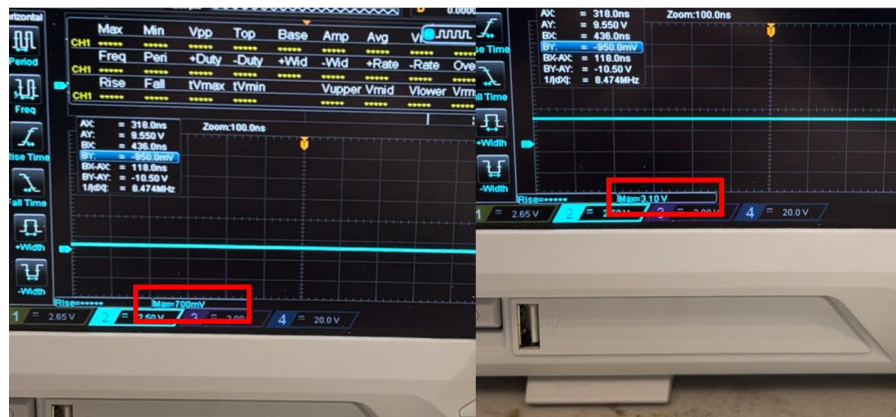


Figure 29: Oscilloscope measuring BPS1 whilst PEN is in the car with a driver.

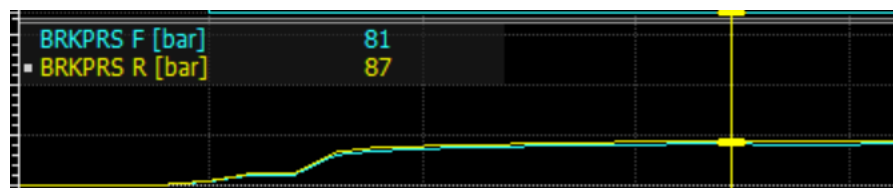


Figure 30: Max brake pressure recorded during competition brake test

```
// BPS Signals
// Raw data [0-1023]
bpsFRaw = analogRead(BPSF_PIN);
bpsRRaw = analogRead(BPSR_PIN);

// Scaled data ~[0 - 2000] - used for brake signal
bpsFScaled = bpsFRaw*2.4438-250; // [psi] assuming 0.5-4.5V maps to 0-2000 PSI Linearly
bpsRScaled = bpsRRaw*2.4438-250; // [psi] assuming 0.5-4.5V maps to 0-2000 PSI Linearly

// Bounded data [0 - 2000] - over/undershoot removed, used for datalogging
bpsFBounded = applyFloorCeil(bpsFScaled, 0, 2000);
bpsRBounded = applyFloorCeil(bpsRScaled, 0, 2000);
```

Equation 1: BRKPRS_F&R Teensy calculation.

The max brake pressure applied during the 2024 FSAE competition brake test was 87 bar (1262 psi) as per figure 13. Using equation 1, bpsRRaw can be back calculated to be 618.7 which corresponds to 2V at the BPS1_SIG_TEENSY analogue input and 3V at BPS1_SIG.

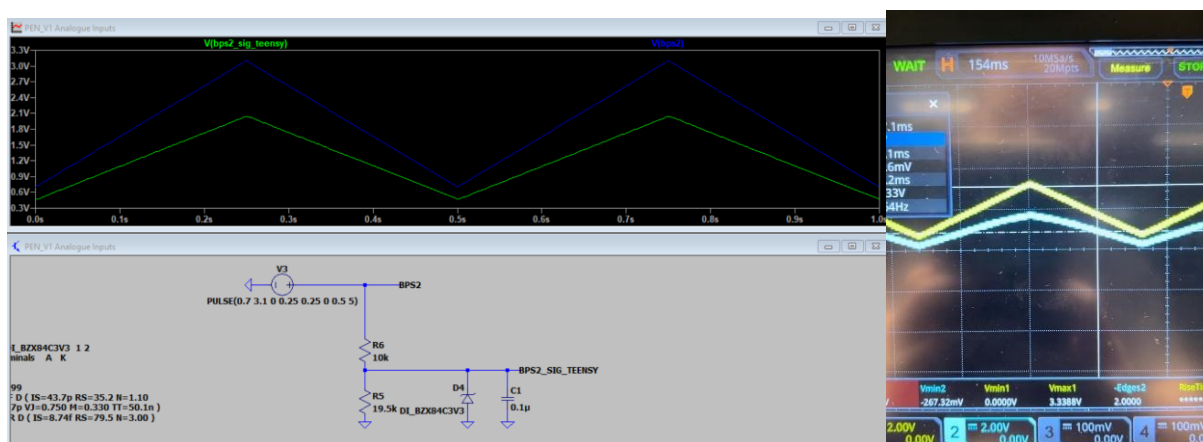


Figure 31: BPSX with 0.1uF capacitor simulation vs bench test (no latency)

Conclusion/ Recommendations

Nonlinearity of TPS1 was caused by the SIG_TEENSY being above $\sim 2.1V$ entering the soft knee zone of the zener diode. Therefore, the zener should be unpopulated however this will leave the teensy unprotected in the event of a voltage surge of the sensor.

TPS2 did not suffer from the non-linearity as the sensor works at a lower voltage range and doesn't encroach on the soft knee. This is possible as the teensy scales the two sensors to have the same range

BPS noise was not replicated or observed with this experiment and the capacitor value was updated. Similar to TPS2, the BPS signals were not affected by the zener due to being $< 2.1V$. When designing the filter for PEN v2, use a similar cut off frequency to the BSPD 1st order filter and account for any voltage drop from the new filter impacting the OCSC values.

PEN PCB and Node Related Rules [6]

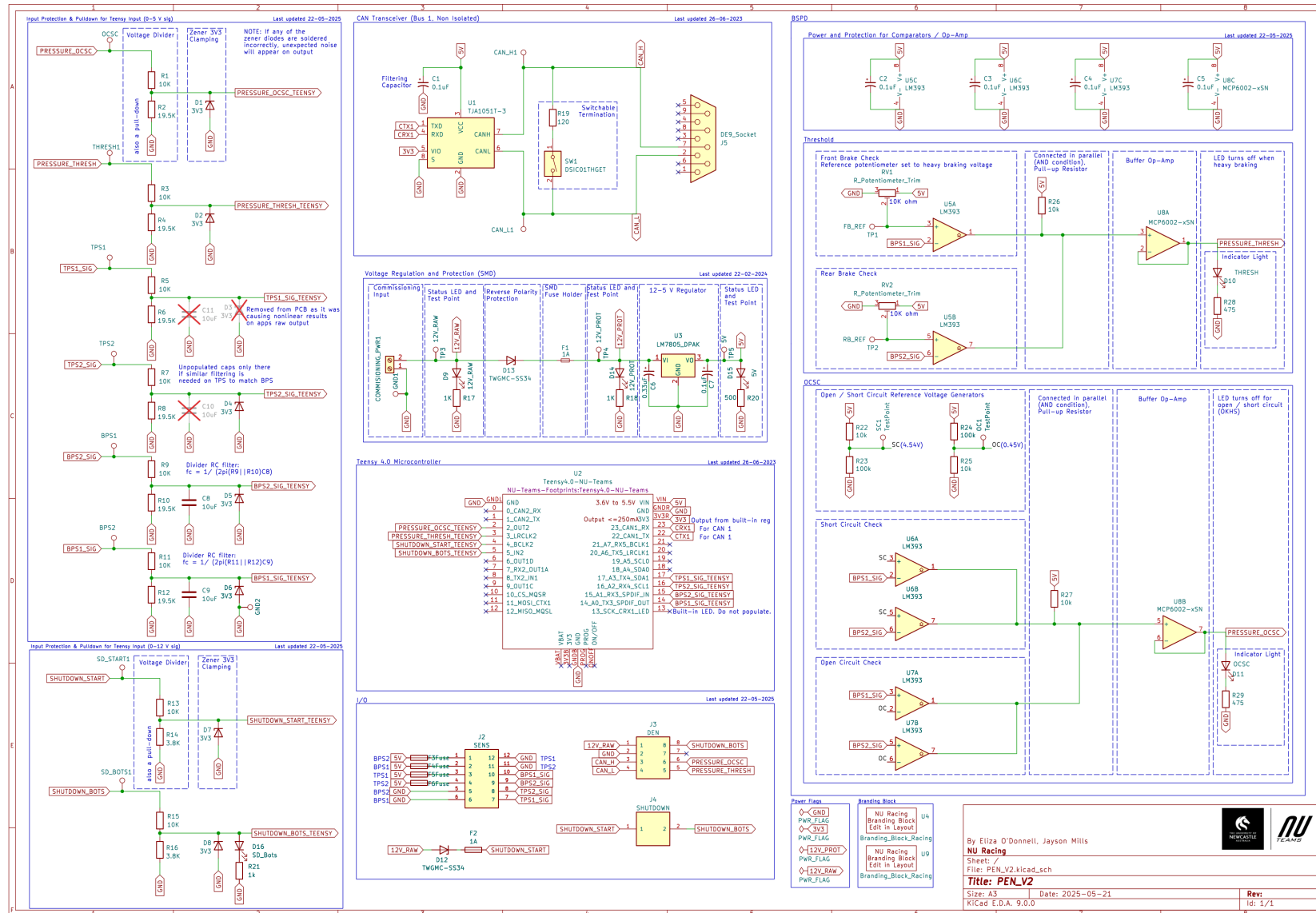
T.3.3	Brake Over Travel Switch - BOTS
T.3.3.1	The vehicle must have a Brake Over Travel Switch (BOTS). Brake pedal travel exceeding the normal range will operate the switch
T.3.3.2	The BOTS must: a. Be a mechanical single pole, single throw (two position) switch (push-pull, push-rotate or flip type) b. Hold if operated to the OFF position
T.3.3.3	Operation of the BOTS to the OFF position must Open the Shutdown Circuit IC.9.2.2 / EV.7.2.2
T.3.3.4	The driver must not be able to reset the BOTS
T.3.3.5	The BOTS must be implemented with analogue components, and not using programmable logic controllers, engine control units, or similar functioning digital controllers.
T.4	ELECTRONIC THROTTLE COMPONENTS
T.4.2	Accelerator Pedal Position Sensor - APPS
T.4.2.1	The Accelerator Pedal must operate the APPS T.1.4.1 a. Two springs must be used to return the foot pedal to 0% Pedal Travel b. Each spring must be capable of returning the pedal to 0% Pedal Travel with the other disconnected. The springs in the APPS are not acceptable pedal return springs.
T.4.2.2	Two or more electrically separate sensors must be used as APPSs. A single OEM type APPS with two completely separate sensors in a single housing is acceptable.
T.4.2.3	The APPS sensors must have different transfer functions which meet one of the two: • Each sensor has different gradients and/or offsets to the other(s). The circuit must have a pull-up or pull-down resistor to bring an open circuit input to 0% Pedal Travel • An OEM pedal sensor with opposite slopes. Non OEM opposite slope sensor configurations require prior approval. The intent is that in a short circuit the APPSs will only agree at 0% Pedal Travel
T.4.2.4	Implausibility is defined as a deviation of more than 10% Pedal Travel between the sensors or other failure as defined in this Section T.4.2. Use of values larger than 10% Pedal Travel require justification in the ETC Systems Form and may not be approved
T.4.2.5	If an Implausibility occurs between the values of the APPSs and persists for more than 100 msec, the power to the (IC) Electronic Throttle / (EV) Motor(s) must be immediately stopped completely. (EV only) It is not necessary to Open the Shutdown Circuit, the motor controller(s) stopping the power to the Motor(s) is sufficient.
T.4.2.6	If three sensors are used, then in the case of an APPS failure, any two sensors that agree within 10% Pedal Travel may be used to define the (IC) throttle position / (EV) torque target and the 3rd APPS may be ignored
T.4.2.7	Each APPS must be able to be checked during Technical Inspection by having one of the two: • A separate detachable connector that enables a check of functions by unplugging it • An inline switchable breakout box available that allows disconnection of each APPS signal.
T.4.2.8	The APPS signals must be sent directly to a controller using an analogue signal or via a digital data transmission bus such as CAN or FlexRay.
T.4.2.9	Any failure of the APPS or APPS wiring must be detectable by the controller and must be treated like an Implausibility, see T.4.2.4 above
T.4.2.10	When an analogue signal is used, the APPS will be considered to have failed when they achieve an open circuit or short circuit condition which generates a signal outside of the normal operating range, for example <0.5 V or >4.5 V. The circuitry used to evaluate the sensor must use pull down or pull up resistors to make sure that open circuit signals result in a failure being detected.
T.4.2.11	When any kind of digital data transmission is used to transmit the APPS signal,

	a. The ETC Systems Form must contain a detailed description of all the potential failure modes that can occur, the strategy that is used to detect these failures and the tests that have been conducted to prove that the detection strategy works. b. The failures to be considered must include but are not limited to the failure of the APPS, APPS signals being out of range, corruption of the message and loss of messages and the associated time outs.
T.4.2.12	The current rules are written to only apply to the APPS (pedal), but the integrity of the torque command signal is important in all stages.
T.4.3	Brake System Encoder - BSE
T.4.3.1	The vehicle must have a sensor or switch to measure brake pedal position or brake system pressure
T.4.3.2	The BSE must be able to be checked during Technical Inspection by having one of: • A separate detachable connector(s) for any BSE signal(s) to the main ECU without affecting any other connections • An inline switchable breakout box available that allows disconnection of each BSE signal(s) to the main ECU without affecting any other connections
T.4.3.3	The BSE or switch signals must be sent directly to a controller using an analogue signal or via a digital data transmission bus such as CAN or FlexRay Any failure of the BSE or BSE wiring that persists more than 100 msec must be detectable by the controller and treated like an implausibility and power to the (IC) electronic throttle / (EV) Motor(s) must be immediately stopped completely. (EV only) It is not necessary to completely deactivate the Tractive System, the motor controller(s) stopping power to the motor(s) is sufficient.
T.4.3.4	When an analogue signal is used, the BSE sensors will be considered to have failed when they achieve an open circuit or short circuit condition which generates a signal outside of the normal operating range, for example <0.5 V or >4.5 V. The circuitry used to evaluate the sensor must use pull down or pull up resistors to make sure that open circuit signals result in a failure being detected.
T.4.3.5	When any kind of digital data transmission is used to transmit the BSE signal: a. The ETC Systems Form must contain a detailed description of all the potential failure modes that can occur, the strategy that is used to detect these failures and the tests that have been conducted to prove that the detection strategy works. b. The failures modes must include but are not limited to the failure of the sensor, sensor signals being out of range, corruption of the message and loss of messages and the associated time outs. c. In all cases a sensor failure must immediately shutdown power to the motor(s).
EV.4.7	APPS / Brake Pedal Plausibility Check
EV.4.7.1	Must monitor for the two conditions: • The mechanical brakes are engaged EV.4.6, T.3.2.4 • The APPS signals more than 25% Pedal Travel EV.4.5
EV.4.7.2	If the two conditions in EV.4.7.1 occur at the same time: a. Power to the Motor(s) must be immediately and completely shut down b. The Motor shut down must stay active until the APPS signals less than 5% Pedal Travel, with or without brake operation The team must be able to demonstrate these actions at Technical Inspection
EV.6.6	Overcurrent Protection
EV.6.6.1	All electrical systems (Low Voltage and High Voltage) must have appropriate Overcurrent Protection/Fusing.
EV.6.6.2	Unless otherwise allowed in the Rules, all Overcurrent Protection devices must: a. Be rated for the highest voltage in the systems they protect. Overcurrent Protection devices used for DC must be rated for DC and must carry a DC

	rating equal to or more than the system voltage b. Have a continuous current rating less than or equal to the continuous current rating of any electrical component that it protects c. Have an interrupt current rating higher than the theoretical short circuit current of the system that it protects
EV.7	SHUTDOWN SYSTEM
EV.7.1	Shutdown Circuit
EV.7.1.5	The Shutdown Buttons, BOTS, TSMS, GLVMS and Interlocks must directly carry the Shutdown Circuit current
EV.7.7	Brake System Plausibility Device - BSPD
EV.7.7.1	The vehicle must have a standalone nonprogrammable circuit to check for simultaneous braking and high power output The BSPD must be provided in addition to the APPS / Brake Pedal Plausibility Check (EV.4.7)
EV.7.7.2	The BSPD must Open the Shutdown Circuit EV.7.2.2 when the two of these exist: Formula SAE® Rules 2025 © 2024 SAE International Page 106 of 143 Version 1.0 31 Aug 2024 • Demand for Hard Braking EV.4.6 • Motor/Accumulator current is at a level where 5 kW of electrical power in the DC circuit is delivered to the Motor(s) at the nominal battery voltage The BSPD may delay opening the shutdown circuit up to 0.5 sec to prevent false trips
EV.7.7.3	The BSPD must Open the Shutdown Circuit EV.7.2.2 when there is an open or short circuit in any sensor input

Pedal Electrical Node

Schematic



The image displays two PCB layouts for the Doom: The Dark Ages car. The top layout is the front side, showing components like the CANBUS module, sensors, and various connectors. The bottom layout is the back side, showing the reverse of the components and additional connectors. The PCB is red with blue traces and features a 'JAM' logo and 'NU RACING' branding.

3D Renders

